

Chapter 1:

This chapter is all about understanding the difference between good and bad hacking and why it matters.

1. The Big Idea: Good vs. Bad Hacking

- Cracking (Bad Hacking): This is breaking into computer systems without permission. It doesn't matter why someone does it—it's always a crime.
- Ethical Hacking (Good Hacking): This is using the same hacking skills, but ****with permission****, to find weaknesses and help protect systems. These are the "good guys."

2. The Types of Hackers (The "Hats")

Think of hackers like characters in an old cowboy movie, identified by the color of their hats:

- White Hat: The Good Guy. They hack with permission to improve security.
- Black Hat: The Bad Guy. They hack illegally to steal, damage, or cause trouble.
- Gray Hat: Someone in the middle. They might break rules but without malicious intent (e.g., hacking a system and then telling the owner about the flaw without asking first).

3. Why Do People Hack?

Hackers are motivated by many things, just like anyone else:

- * Curiosity and the love of solving puzzles
- * Wanting fame or recognition
- * Revenge
- * To make money
- * Patriotism or political reasons

4. A Quick History of Hacking

- 1950s/60s: Started at universities (like MIT) as a smart, playful way to explore computers.
- 1970s: "Phone phreaks" figured out how to make free phone calls.
- First password hacks were a response to the Compatible Time Sharing System (CTSS)—Developed in the early 1960s
- 1980s: Personal computers became common, and hacker groups grew. Bad software (viruses, worms) started appearing. **War dialers** were developed to search for open modems
- Today: The term "hacker" often has a negative meaning because of the actions of malicious "crackers."

5. Why Are Ethics So Important?

- * Without clear rules of conduct (ethics), it's hard for security professionals to be trusted.
- * The skills of a good hacker and a bad hacker are identical. The only difference is permission and intent.
- * The industry is working on creating standard certifications and codes of ethics to help everyone tell the good guys from the bad guys.

6. What Needs to Be Protected?

- Our Data: Personal information and private files are a top priority to secure.
- Resources: Using a company's computer power or internet without permission is theft, even if it seems harmless.
- Software: Copying paid software without a license is also a crime.

In One Sentence:

Ethical hacking is a necessary and legal profession that uses hacking skills to defend and protect systems, while cracking is the illegal and harmful use of those same skills.

Chapter 2:

Reconnaissance - The "Scouting" Phase

Think of reconnaissance like a thief "casing" a neighborhood before a robbery. In hacking, it's the first step where an attacker gathers as much information as possible about a target to find the best way to break in.

➤ Part 1: It's Not Always Illegal (But It Can Be)

Reconnaissance is just finding your target and figuring out how to attack it. A lot of this is totally legal and happens every day.

Legal Examples: Looking up a company online, calling their customer service, taking a public tour of their building, or making friends with an employee.

Questionable (Ethically Gray) Examples: Reading papers on someone's desk, picking up trash in the parking lot to look for info, or "wardriving" (driving around to find wireless networks).

Illegal Examples: Posing as a fake company to scam people, breaking into an office to look for data, or stealing someone's garbage.

➤ Part 2: Social Engineering - Hacking People, Not Computers

This is the art of manipulating people into giving up secrets or access. It works because people are naturally helpful and trusting.

Common Tricks:

- Impersonation: Pretending to be someone else, like an IT technician or a delivery person.
- Bribery/Blackmail: Offering money or threatening someone to get what they want.
- Deception: Getting a job at the target company to steal from the inside.
- Conformity—Hacker convinces the victim that they have a lot in common and that they share the same values. Hacker becomes the victim's good friend
- Reverse social engineering—Hacker projects herself as an authority vested with the power to solve peoples' problems
- Phishing: Sending fake emails that look real to trick you into giving up passwords or personal info (e.g., a fake email from your bank).

How to Protect Yourself:

- Be suspicious. Don't give information to people you don't know.
- Never share passwords over the phone or email.

- Lock your computer when you step away.
- Verify identities. If someone asks for sensitive info, check if they are who they say they are.

➤ Part 3: Dumpster Diving - One Person's Trash is a Hacker's Treasure

Hackers literally go through a company's trash to find valuable information.

What they look for:

- * Old paperwork with passwords, client lists, or internal memos.
- * Old hardware like hard drives or USB sticks that still have data on them.
- * Company directories or policy manuals.

How to Stop It:

- *SHRED EVERYTHING! Use a cross-cut shredder for all documents.
- *Have a clear policy for handling trash.
- *Destroy old data storage devices completely: (don't just throw them out or break CDs in half—they need to be shredded or melted).

➤ Part 4: Internet Footprinting - The Digital Detective Work

This is the technical side of reconnaissance. Hackers love it because they can do it from anywhere, and it's completely legal.

There are four main ways they do it:

1. Web Searching:

- Using **Google** to find info about a company, its employees, and its systems.
- Looking at the **HTML source code** of a website for hidden comments or clues.
- Reading **newsgroups and forums** where employees might accidentally discuss work problems.

2. Network Enumeration:

- Using tools like **WHOIS** to find out who owns a website (domain name) and their contact information. It's like looking up a license plate.

3. DNS-Based Reconnaissance:

The **Domain Name System (DNS)** is like the internet's phonebook. Hackers use commands like `nslookup`, `dig`, and `host` to:

- * Find the IP addresses of a company's servers.
- * Try a **"zone transfer"**—a trick to get a full list of all a company's internal computer names (if the server is misconfigured).

4. Network-Based Reconnaissance:

- `ping`: A simple command to see if a computer is online and responding.
- `tracert`(or `tracert` on Windows): Maps the path your connection takes to a website, revealing what servers and routers a company uses.

- netstat: A command that shows all active connections on a computer, showing what other systems it's talking to.

Key Takeaway:

Reconnaissance is all about information gathering. The more an attacker knows about their target (the people, the trash, the digital footprint), the easier it is to find a weakness to exploit. The best defense is to be aware of what information you and your company are giving away, both online and offline.

Chapter 3:

Scanning Tools: Imagine you're a security guard checking all the doors and windows of a building to see which ones are unlocked. **Scanning tools** are the digital version of that for computers and networks. They automatically check other computers on a network to find weak spots (vulnerabilities) or open entry points (ports) that a hacker could use to get in.

1. What is a Scanner?

- * It's a software tool that probes other computers (hosts) on a network.
- * Its main job is to find and report on open ports (like digital doors) and known security weaknesses.
- * Port Scanner: A common type of scanner that checks if a port is open (accepting connections) or closed and can often identify what software is listening on that port.

2. A Quick History Lesson (Evolution)

- * Before the Internet: The idea started with tools to monitor connections between big computers (mainframes) and simple terminals.
- * War Dialers: In the early days, hackers used scripts called "war dialers" that would automatically dial a range of phone numbers to find which ones were connected to a computer modem.
- * Shell Scripts: System administrators started writing their own simple scripts to check their networks for weaknesses.
- * Today: Scanners are now powerful, easy-to-use programs available for all major operating systems.

3. How Do They Work? (In Simple Terms)

Scanners are not smart AI; they just automate a repetitive process. They:

- Connect** to a target computer.
- Check** what services (like a web server or email server) are running.
- Compare** those services against a database of known vulnerabilities.
- Report** what they find.

4. Types of Scanning (The Different "Approaches")

Scanners use different methods to check ports without getting caught.

- * TCP Connect Scan: The most basic method. It's like walking up to a door and checking the handle. It's easy to detect but doesn't require special privileges.

- * Half-Open (SYN) Scan: A sneakier method. It starts to open a connection but stops halfway through to avoid being logged. It's harder to detect but needs higher user privileges.
- * UDP Scan: Checks a different type of port (UDP), which is used for things like video streaming or DNS. These are slower to scan because they often don't respond if closed.
- * Ping Scan: The simplest check. It just asks the computer "Are you there?" (sends an ICMP echo request). If it gets a reply, the host is online.
- * Stealth Scanning: A general term for any sneaky technique used to scan a host without the target's logging systems noticing, especially useful for scanning through firewalls.

5. Popular Scanning Tools (The "Big Names")

- * Nessus: A very powerful and popular ****vulnerability scanner****. It uses a database of thousands of known weaknesses (plug-ins) to test a system. It has a client/server architecture.
- * Nmap ("Network Mapper"): The best all-around network scanner. It's famous for being:
 - * Free, powerful, and flexible. It can do almost every type of scan, identify operating systems, and has both a command-line and graphical interface. Example command: `nmap -A 192.168.0.1` does an intense scan of that IP address.

- * SARA & SAINT: Third-generation tools designed for security professionals to analyze networks. They are more advanced and are meant to work with other security tools.
- * Strobe: A very fast but old TCP port scanner that finds all open ports on a target.
- * Cheops: An older GUI-based tool for Linux. (Note: The slides mention it can be very slow and use a lot of CPU).

Key Takeaways / Summary

- * Purpose: Scanners are used by both good guys (admins securing their networks) and bad guys (hackers looking for a way in) to find weaknesses in computer systems.
- * They Automate a Process: They automatically check for **open ports** and **known vulnerabilities**.
- * They Are Not Magic: They can only find problems that are already known and listed in their databases. They can't find brand-new, unknown vulnerabilities.
- * Be Careful: Scanning networks that you don't own can get you in trouble and might get you blocked by your Internet Service Provider (ISP), as it can look like an attack.

In one sentence: Scanning tools are automated programs that check other computers for open doors and known weak spots, helping to secure a network or, if used maliciously, break into it.

Chapter 4:

What is a Sniffer? (The Big Idea)

Imagine a sniffer is like a ****wiretap for computer networks****. It's a program that can secretly listen to all the data (called "packets") traveling across a network, just like someone listening to every conversation on a phone line.

1. Types of Sniffers

There are three main kinds

- * **Bundled:** Free tools that come with your operating system.
Examples: Network Monitor (Windows), tcpdump (Linux)
- * **Commercial:** Powerful tools you pay for. They often have nice interfaces and customer support. Examples: Sniffer Pro, EtherPeek NX
- * **Free:** Very capable tools you can download for free, but you're on your own for help. Examples: Wireshark (the most famous one), Snort

2. How Do They Work?

Normally, your computer's network card only pays attention to data specifically sent to it.

- * A sniffer puts the card into "promiscuous mode." This is like telling it to ****eavesdrop on every single conversation**** on the network, not just the ones for you.
- * It captures this data, filters it, and saves it so someone can look through it later.

Components of a Sniffer

- **Hardware:** NIC is the hardware most needed

- Capture Driver:Captures the network traffic from the Ethernet connection.Filters out the information that you don't want And then stores the filtered traffic information in a buffer
- Buffer:Dynamic area of RAM that holds specified dataMethods of storing captured data.Stored until the buffer is full with information.*Round-robin method
- Decoder–Interprets binary information and then displays it in a readable format
 - Packet Analysis–Sniffers usually provide real-time analysis of captured packets

3. Where are They Placed?

To be effective, a sniffer needs to be placed where it can "hear" the most interesting traffic. Good spots include:

- * Near important servers (like password servers).
- * On network connections that lead to the internet.
- * On routers or key cable points.
- * Media Access Control (MAC) Addresses

4. How to Detect a Sniffer (Very Hard!)

Since sniffers just listen and don't transmit, they are very hard to find. But you can try to:

- * Check for promiscuous mode: Use tools like `AntiSniff` or run the command `ifconfig -a` on a computer to see if its network card is in listening mode.
- * Look for suspicious activity: See if a computer is doing unexpected DNS lookups or seems slower (working harder to listen to everything).

5. How to Protect Yourself from Sniffers

You can't always stop someone from listening, but you can make what they hear **useless**. The solution is **encryption**.

- * For Web Browsing (HTTPS): Use websites that start with `https://`.

This uses **SSL/TLS** encryption.

- * For Email: Use encrypted email standards like **PGP** or **S/MIME**.

- * For Remote Logins: Use **SSH** instead of Telnet. SSH encrypts the entire connection.

- * For General Traffic: Use **IPSec** to encrypt all data between devices.

The golden rule: If your data is encrypted, a sniffer might capture it, but they won't be able to read it.

- Key Takeaways:

- *A sniffer* is a network eavesdropping tool.

- * It works by making a network card listen to all traffic(promiscuous mode).

- * They are very hard to detect because they are passive.

- * The best defense is encryption (SSL, SSH, PGP). If the data is encrypted, the sniffer can't understand it.

Think of it like sending a letter. A sniffer can intercept the envelope, but if you wrote the message in a secret code (encryption), the interceptor can't read it

NB: learn sniffer programs from slides

Chapter 5:

What is TCP/IP? (The "What")

Think of **TCP/IP** as the **rulebook or language** that computers use to talk to each other on the internet. It's not one single thing, but a whole set of rules that ensures data gets from point A to point B correctly.

- * IP (Internet Protocol): This is like the **postal system**. Its job is to address the package (the data packet) and get it to the right neighborhood (the right computer network). It doesn't care if the package gets lost—it just focuses on delivery. When a large IP packet is sent over a network, it is broken down. Called fragmentation
- * TCP (Transmission Control Protocol): This is like a **reliable courier** working for the postal system. It makes sure the package was actually received, puts the packages in the right order if they arrive out of sequence, and re-sends any that got lost along the way. Unauthorized users may launch a denial-of-service attack on the destination computer

Data encapsulation

- Enclosing higher-level protocol information in lower-level protocol information
- Also called data hiding

- **How Do Computers Connect? (The "How")**

They use a **Three-Way Handshake**. Imagine two people meeting:

1. SYN (Synchronize): Computer A says to Computer B, "Hello, can we talk?"

2. SYN-ACK (Synchronize-Acknowledge): Computer B replies, "Sure, let's talk! Ready when you are."

3. ACK (Acknowledge): Computer A says, "Great, here's my first piece of data."

Now the connection is established, and they can chat (exchange data).

- The Big Problem: Vulnerabilities (The "Weaknesses")

When the rules of TCP/IP were written, security wasn't a big priority. Bad guys can exploit these old rules. Here are the main attacks:

1. IP Spoofing: Faking your return address. *Like: Sending a prank letter but putting your enemy's address as the return address. The response goes to them, not you.

2. TCP SYN Flood Attack: Overwhelming the handshake.* Like: A thousand people simultaneously saying "Hello, can we talk?" to a single shopkeeper. The shopkeeper gets so busy saying "Sure!" that they can't help any real customers. This is a **Denial-of-Service (DoS)** attack.

3. Connection Hijacking: Jumping into an existing conversation.*Like: Eavesdropping on a phone call, then mimicking one person's voice to take over the conversation.

4. ICMP Attacks:Abusing the internet's "ping" system.

* Like:Yelling "fire" in a crowded theater to cause chaos. These packets can be used to trick computers or disrupt connections.

5. RIP Attacks: Tricking the internet's road signs. * Like: Changing road signs so that all the traffic (data) gets sent to the wrong destination (the hacker's computer).

- How Do We Protect Ourselves? (The "Solution")

We can't change the core rules of the internet, but we can build defenses around them:

*Use Firewalls: Act as security guards that check incoming and outgoing traffic, blocking anything that looks suspicious (like spoofed packets).

*Change Default Settings: Make it harder for attackers by changing things like how long a system waits for a handshake to finish.

*Use Encryption: This is the biggest one! If you scramble the data, even if a hacker intercepts it, they can't read it.

* VPNs (Virtual Private Networks): create an encrypted "tunnel" for your data to travel through.

* IPSec: is a built-in set of rules that automatically encrypts and authenticates all the data being sent between two points on a network.

Super Simple Summary

* TCP/IP: is the language of the internet.

* The *Three-Way Handshake* is how computers start a conversation.

* The main weakness is that this language *wasn't built with security in mind*.

* Hackers can spoof addresses, flood connections, and hijack sessions.

- We protect ourselves with firewalls, better settings, and encryption (like VPNs and IPSec)

Chapter 6:

Cryptography Basics

- Cryptography: Process of converting plaintext to ciphertext (encryption) and back (decryption).
- Transposition Ciphers: Rearrange letters or words; vulnerable to frequency analysis.
- Substitution Ciphers: Replace letters/groups with others (e.g., Enigma machine).
- Key Terms: Cleartext, ciphertext, key, algorithm, hash.

Types of Encryption

- Symmetric Key Encryption
- Same key used for encryption and decryption.
- Faster than asymmetric encryption.

Examples:

*Stream Ciphers: Use key streams (e.g., one-time pads, XOR operations).

*Block Ciphers: Operate on fixed-size blocks (e.g., DES, AES).

- Asymmetric Key Encryption
- Uses public and private keys.
- Public key encrypts; private key decrypts.

Examples: RSA, Diffie-Hellman, DSS, Elliptic Curve Cryptography.

- DSA (Digital Signature Algorithm)
 - Digital signature connects documents with the holder of a specific key
 - Considered too slow for general encryption
- Digital Time Stamps
 - Connects document with a specific time of origination
- Cryptanalyst decodes messages to make them readable
 - First and most important step in cryptanalysis
 - Detecting the key values

Common Ciphers

- Symmetric Ciphers
 - DES: 56-bit key, 64-bit blocks.
 - 3DES: Triple encryption with DES.
 - AES (Rijndael): 128–256-bit keys, U.S. government standard.
 - IDEA: 128-bit key, 64-bit blocks.
 - RC4: Stream cipher, fast but moderate security.
 - Blowfish: Fast, designed by Bruce Schneier.
- Asymmetric Ciphers
 - RSA: Based on factoring large primes.
 - Diffie-Hellman: Secure key exchange.
 - DSS: Digital signatures.
 - Elliptic Curve: Shorter keys, high security.
 - Lattice-Based: Theoretical, not widely used.

Cryptographic Hash Functions

- Convert variable-length input to fixed-length hash.

Examples:

- MD5: 128-bit hash, now considered weak.
- SHA family: SHA-1 (160-bit), SHA-2 (up to 512-bit), more secure.

Password Attacks

- Dictionary Attacks: Use common words; hybrid attacks add numbers/symbols.
- Brute-Force Attacks: Try all possible combinations.
- Observation: Shoulder-surfing, eavesdropping.
- Keylogger: Record keystrokes.
- Social Engineering: Manipulate users into revealing passwords.
- Sniffing: Capture unencrypted passwords over networks.
- Password File Theft: Steal hash files for offline cracking.

Password-Cracking Tools

- Cain and Abel: Windows-based, multi-purpose.
- Crack: UNIX password cracker.
- John the Ripper: Fast, multi-platform.
- Telnet_Crack: Targets Telnet passwords.
- THC Hydra: Network authentication cracker.
- L0phtCrack: Windows password auditor (discontinued).

Best Practices for Password Security

- Use strong, unique passwords.
- Avoid personal names, common words.
- Include random characters, numbers, symbols.
- Don't write passwords down.
- Use multi-factor authentication where possible.

Chapter 7:

What is Spoofing? (The Big Idea)

Imagine someone calling your phone, but they disguise their number to look like your bank's number. That's ****spoofing**** in the digital world. It's the act of ****faking an identity**** to trick a computer or a person into trusting you.

- The Core Problem: Trust & Authentication

* Trust: Computers often automatically trust each other based on their digital address (like an IP address).

* Authentication: This is how a computer proves its identity is real.

* The Flaw: These two have an ****inverse relationship****. The more two computers trust each other, the less they check each other's identity. Spoofing exploits this lazy trust.

- How Does an IP Spoofing Attack Work? (The 5 Steps)

An attacker pretending to be a trusted computer must:

1. Find Targets: Find two computers (Server A and Server B) that trust each other.

2. Knock Out the Impersonated Computer: Overload the real trusted computer (Server A) with junk traffic so it can't respond (this is called a DoS or SYN flood attack). It's like putting it to sleep.

3. Pretend to Be It: Send messages to the target computer (Server B) using the sleeping computer's (Server A's) fake address.

4. Guess the Secret Handshake: Accurately predict the complex number sequence (ISN) that Server B expects to see to prove the connection is real. This is the hardest part.

5. Get In & Clean Up: If the guess is right, the attacker is in! They then wake up the sleeping computer so no one gets suspicious.

- Types of Spoofing (The Different Flavors)

*IP Spoofing: Faking a computer's IP address.

*ARP Spoofing: Tricking your local network into sending data to the hacker's computer instead of the real router.

*Web Spoofing: Creating a fake website that looks identical to a real one (like a fake bank login page) to steal your info.

*DNS Spoofing: Tricking the internet's "phonebook" (DNS) into directing you to a malicious website when you type a real web address.

*Blind Spoofing: The hacker is shooting in the dark. They can only see the target they are attacking, not the responses. It's harder because they have to guess everything correctly. (Common from the internet).

*Active Spoofing: The hacker is eavesdropping on the conversation. They can see both parties and interfere in real-time. This is much easier for the hacker. (Common on local networks like WiFi).

Costs of Spoofing

If successful, spoofing can lead to:

- * Economic Loss: Theft of money or valuable data.
- * Strategic Loss: Competitors stealing future business plans.
- * Data Loss: General loss of private company information.

Common Spoofing Tools (Hacker's Toolkit)

- * Apsend: Used to create floods of fake data (SYN flood, Ping flood) to overwhelm a target.
- * Ettercap: A powerful, all-in-one tool for ARP poisoning and sniffing data on a network.
- * Arpspoof: A simple tool specifically for tricking devices on a local network (ARP spoofing).

How to Defend Against Spoofing (The Good Guy's Guide)

- To stop IP Spoofing:
 - * Don't use trust based only on IP addresses. Use stronger passwords and keys.
 - * Use a firewall to filter and check incoming traffic.
 - * Use encryption (like IPSec) so even if data is intercepted, it can't be read.
- To stop ARP Spoofing/Poisoning:
 - * Use ****static ARP tables**** (a manual list of trusted devices that can't be changed by hackers).

* Log and monitor any changes to the network's ARP tables.

NB: you are at your own risk not adding the slides learning this

In a Nutshell:

****Spoofing is digital impersonation.**** It works because computers are too trusting. You can fight it by using stronger authentication, encryption, and by monitoring your network for suspicious changes.

Chapter 8:

Session Hijacking:

Imagine two computers (let's call them ****Alice**** and ****Bob****) are having a trusted, private conversation. **Session hijacking** is when a hacker (let's call them ****Eve****) secretly jumps into the middle of that conversation and pretends to be one of them, stealing control.

1. How Does It Work? (The Hacker's View)

* The Goal: Eve wants to sneak into the conversation **after** Alice and Bob have finished logging in and verifying who they are (authentication).

*The Problem: Computers keep track of their conversation by numbering their messages. If Eve just starts sending messages as Bob, Alice will get confused because the message numbers will be wrong.

- The Main Tactic: Eve has a few ways to do this:

*Man-in-the-Middle: Eve positions herself between Alice and Bob, secretly listening to and controlling all their messages (like a malicious operator on a phone line).

*Blind Hijacking: Eve isn't in the middle but just guesses what the next message number should be and sends a fake command. It's less reliable.

*ACK Storm: When Eve hijacks the session, the real computer (say, Bob) keeps sending messages that Alice doesn't understand. Alice replies with error messages ("ACK packets"), creating a storm of traffic that can give the attack away.

2. Two Key Methods Hackers Use

To solve the "ACK Storm" problem and gain full control, hackers use these tricks:

- Route Table Modification

* Every computer has a **map** (a route table) that tells it how to send data to other computers.

* Eve **hacks this map** and tells Alice: "The fastest way to talk to Bob is through me!" Now, all of Alice's messages for Bob get sent directly to Eve first.

- ARP Attacks

* Computers have two addresses: a logical name (IP address) and a physical, unique serial number (MAC address).

* When Alice wants to talk to Bob, she asks out loud, "What is Bob's MAC address?"

* Eve **yells the answer first**, lying: "Bob's MAC address is [Eve's MAC address]!" Alice believes the lie and sends all future messages for Bob directly to Eve.

3. TCP vs. UDP Hijacking

* TCP Hijacking: Like a reliable registered mail service. It keeps track of every package. Hijacking it is harder because you have to guess or control the sequence numbers.

* UDP Hijacking: Like yelling a message across a crowded room. There's no confirmation it was received. Hijacking is *much easier* because the hacker just has to shout a fake message before the real person can respond. There's no built-in tracking to stop it.

4. A Famous Tool: Hunt*

`Hunt` is a popular software tool that makes it easy for hackers to:

- * List all active connections on a network.
- * Watch conversations.
- * Hijack a session using the ARP attack method described above.
- * Synchronize the connection to avoid detection.

5. How to Defend Against It

You can protect yourself with two main strategies:

- Use Encryption (The Best Defense!)

* If Alice and Bob are speaking in a secret code (encryption) that only they understand, it doesn't matter if Eve intercepts the messages. She won't be able to read them or fake them.

Always use encrypted protocols like SSH (instead of Telnet), HTTPS (instead of HTTP), and VPNs. Avoid plaintext protocols like Telnet, FTP, and basic POP3.

- Storm Watching

- * Use security systems (Intrusion Detection Systems - IDS) to watch the network.

- * Tell the system to sound an alarm if it suddenly sees a huge "storm" of error messages (ACK packets), as this is a classic sign of a hijacking in progress.